When Bagging Beats Boosting: Single Trees, Bootstrap Aggregation, and SAMME on Raw Olivetti Faces

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Abstract

We study closed-set face identification from raw 64×64 *Olivetti images under a frozen, roll-number-seeded split of six training, two validation and two test images per identity. We compare a single CART tree, bootstrap aggregation (bagging) with up to 200 trees, and a from-scratch SAMME multiclass AdaBoost. Every choice is made on clean validation accuracy; a centred* 16×16 *block is zeroed only at test time. The selected single tree overfits (training accuracy 1.0000, 41 leaves) and reaches 0.6000 validation and 0.5000 test accuracy. Bagging with* $B = 200$ *reaches 0.9250 validation and 0.8375 clean test accuracy, and retains 0.7375 when the central block is zeroed. SAMME with depth 4 and 50 rounds reaches 0.4375 validation and 0.3875 test accuracy, and 0.2750 masked. Thirty within-identity bootstrap resamples show that bagging cuts measured variance by a factor of 30.9, while squared bias rises from 0.4208 to 0.5270; the variance reduction dominates and refit accuracy improves from 0.4687 to 0.7413. We report the unsuccessful and ambiguous outcomes as well: boosting's hard-vote decisions remain weak on 240 training faces, and in a masked case where bagging still gives the true identity 49 of 200 votes, boosting's normalised score for that identity is only 0.1484.*

1. Introduction

Face identification from raw pixels is a deliberately unfriendly test bed for classical classifiers. Each image is a 4096-dimensional vector, only 240 images are available for training, and the forty identities are separated by subtle lighting, pose and expression changes. A decision tree can fit this training set perfectly by isolating individual faces, but such a tree has no reason to generalise. Ensembling is the natural response: bagging attacks variance by averaging trees fitted to bootstrap resamples, while boosting attacks bias by reweighting examples that the current weak learners get wrong.

This report answers the three assignment questions under one fixed protocol. The split is generated once from the seed obtained by reading every numeric character of the roll number `P26AI0006` in order, namely 260006, and is reused by every experiment. It assigns six training, two validation and two test images to each identity. Model selection uses clean validation accuracy only; the clean test set is touched only for final reporting, and the masked test set is never used for training or selection. No PCA or other dimensionality reduction is applied.

Our contributions are: (i) an exact, saved split and a deterministic Q1–Q3 pipeline with artifact checks; (ii) student implementations of bagging and SAMME, including explicit rejection of random learners; (iii) a diagnostic study of OOB error, disagreement, margins, weight concentration and masked behaviour; and (iv) a measured bias–variance decomposition over 30 resamples. The headline is that bagging wins clearly, while SAMME is stable but weak: its weighted training error falls from 0.8500 to 0.6143, yet its test accuracy is 0.3875, below the single tree's 0.5000.

2. Protocol and Reproducibility

2.1. Data and frozen split

We use `sklearn.datasets.fetch_olivetti_faces` [9]: 40 identities, ten images each, 64×64 pixels, values in $[0, 1]$. The cache is stored as `data/olivetti_faces.npz`; its image array hashes to `e6b9b0fe62f6`. The seed rule gives

$$\text{seed} = \text{all digits of } \texttt{P26AI0006} \text{ in order} = 260006. \quad (1)$$

A single `numpy.random.default_rng(260006)` stream is consumed identity by identity in ascending or-

der. For each identity the ten dataset indices are permuted once; the first six become training images, the next two validation images and the last two test images. The resulting 240/80/80 partition has SHA-256 `8903d5baad04` and is saved to `results/split.npz` and `results/split.csv`. Every script loads this file; the split is never regenerated inside an experiment.

2.2. Masked test set and selection discipline

For the masked condition we copy the test matrix and set rows 24–39 and columns 24–39 of every image to zero, a centred 16×16 block. All other pixels are unchanged; training and validation images remain clean. The mask is applied only when evaluating a fitted model, so this is a response test, not augmentation.

Every hyperparameter and ensemble size is selected by clean validation accuracy. Ties are resolved by the first row in a fixed grid order (shallower depth, then smaller leaf, then smaller B or fewer rounds). The clean test and masked test columns are reported after selection; neither enters a selection rule.

2.3. Implementation

`scikit-learn`'s [8] `DecisionTreeClassifier` is the only base learner, as permitted. Bagging and SAMME are implemented in `src/common.py`. Bagging draws one nested sequence of 200 bootstrap replicates and 200 tree seeds; an ensemble of size B is exactly the first B trees of the 200-tree ensemble, making the B-curve paired rather than a collection of independently seeded runs. SAMME fits each tree with the current sample weights, records weighted error, α_t and the weight distribution, and stops explicitly if a learner is no better than random. The full pipeline is `python -m src.run_all`; the run log, resolved configuration, pip freeze and SHA-256 manifests are stored under `results/`. The run took 1.34 hours.

3. Q1: Single Trees and Bagging

3.1. Decision-tree grid

The grid crosses maximum depth $\{2, 4, 8, 16, \text{unrestricted}\}$ with minimum leaf size $\{1, 2, 4\}$, giving fifteen trees. Figure 1 plots validation accuracy and the selected leaf size. Accuracy rises monotonically with depth; shallow trees underfit badly (depth 2 reaches 0.0750 validation, depth 16 reaches 0.2875), and the unrestricted tree is the only one above 0.5. The selected configuration is unrestricted depth with `min_samples_leaf`$= 1$: training 1.0000, validation 0.6000, test 0.5000. It has 41 leaves, 5.85 training samples per leaf on average, and a singleton-leaf fraction of 0.024. Perfect training accuracy on 240 samples is memorisation, not a model of faces. The complete grid is in `results/tree_grid.csv`.

3.2. Bagging and the ensemble-size curve

Bagging [1, 2] fits B trees to bootstrap resamples of the 240 training images and predicts by plurality vote. Table 1 and Figure 2 report $B \in \{1, 5, 10, 25, 50, 100, 200\}$. A single bootstrap tree ($B = 1$) reaches 0.4625 validation and 0.3875 test accuracy, worse than the tree fitted to all training data. Averaging fixes this quickly: $B = 10$ gives 0.7625 validation, $B = 50$ gives 0.8500, and $B = 200$ gives 0.9250 validation and 0.8375 test accuracy. Training accuracy is already 1.0000 at $B = 10$; further trees reduce variance, not training error. $B = 200$ is selected for its highest validation accuracy; $B = 100$ is close (0.9000 validation) and reaches the same 0.8375 test accuracy.

3.3. Out-of-bag error, disagreement and margin

Out-of-bag accuracy at $B = 200$ is 0.8042, close to the 0.8375 test estimate and well above the single tree's 0.5000. Coverage is 1.0: every training image is out of bag for at least one tree. The mean number of distinct training indices per bootstrap sample is 151.90, against the theoretical 151.89, confirming with-replacement sampling.

Trees disagree on 0.7754 of validation pairs on average. This is healthy diversity: if every tree made the same error, averaging could not help. The mean winning-minus-runner-up vote margin is 0.3023, so the winner holds about 30 percentage points more votes than the runner-up. The relationship is not monotone in B (0.2675 at $B = 5$ to 0.3023 at $B = 200$) because adding trees can move both the winner and the runner-up. OOB accuracy and margin improve with B, while disagreement jumps to about 0.78 by $B = 5$ and then stays almost flat. Disagreement measures diversity, not correctness; OOB accuracy and margin track accuracy.

3.4. Clean versus masked test, and face cases

The masked column of Table 1 is the most striking Q1 result. A single bootstrap tree falls to 0.0250 masked accuracy: it has learned splits on the central pixels and has no redundancy when they are removed. Five trees already recover 0.4125; 200 trees reach 0.7375, only 0.1000 below their clean accuracy. Averaging makes the prediction robust to destroying one local region even though individual trees are not region-invariant.

Confidence reinforces the difference. On clean images the selected tree's mean winning confidence is 1.0000 and bagging's is 0.3934; under the mask they are 1.0000 and 0.2657. The tree is maximally confident in both conditions, including when wrong, while bagging's confidence falls with its accuracy. Vote fractions are not calibrated probabilities, but they are sensitive to lost evidence.

Figure 3 shows two faces where bagging corrects the tree and two where both fail. For dataset index 376 (true 37) the tree predicts 3 with probability 1.0; bagging gives 37 118 of 200 votes, a margin of 0.500. For index 175 (true

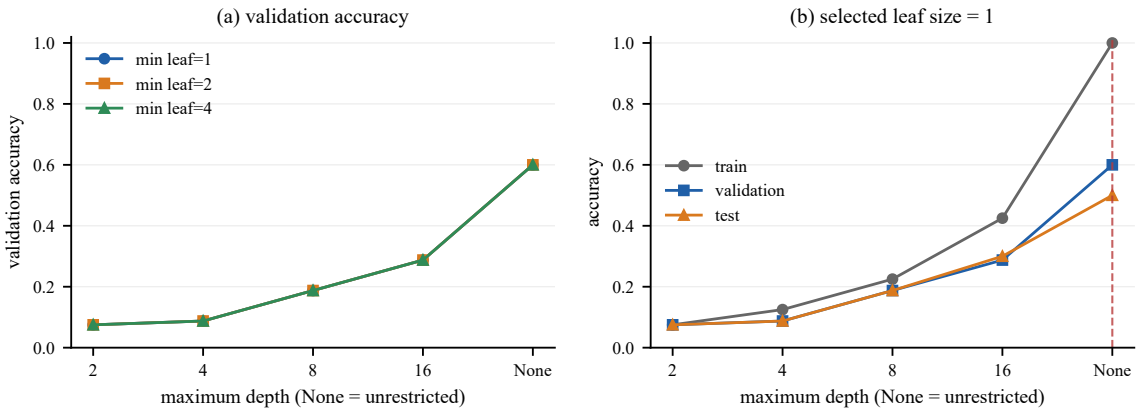


Figure 1. Q1 tree grid. (a) Validation accuracy for the fifteen depth × leaf-size combinations; unrestricted depth is the only setting above 0.5. (b) Train, validation and test accuracy at the selected leaf size 1. Training reaches 1.0000 while validation stalls at 0.6000.

Table 1. Bagging with the frozen tree setting. All B values use the same nested bootstrap sequence; OOB accuracy is computed only on covered training samples. Disagreement and margin are on the clean validation set. "Distinct" is the mean number of distinct training indices per bootstrap sample; the theoretical value $n(1-(1-1/n)^n)$ is in parentheses.

B	Train	Val	Test	Masked	OOB	Disagreement	Margin	Distinct (theory)
1	0.7167	0.4625	0.3875	0.0250	0.2766	0.0000	1.0000	146.0 (151.9)
5	0.9708	0.6625	0.5500	0.4125	0.3146	0.7787	0.2675	151.8 (151.9)
10	1.0000	0.7625	0.6375	0.5000	0.4008	0.7633	0.2738	153.1 (151.9)
25	1.0000	0.8250	0.7375	0.4500	0.6042	0.7741	0.2895	151.6 (151.9)
50	1.0000	0.8500	0.7875	0.6625	0.7250	0.7776	0.2887	151.5 (151.9)
100	1.0000	0.9000	0.8375	0.6625	0.7708	0.7779	0.2928	151.7 (151.9)
200	1.0000	0.9250	0.8375	0.7375	0.8042	0.7754	0.3023	151.9 (151.9)

17) the tree predicts 22; bagging gives 17 114 votes, margin 0.475. Among shared errors, index 2 (true 0) is predicted 26 by the tree and 15 by bagging (56 votes, margin 0.115), while index 154 (true 15) is predicted 5 by both (52 votes, margin 0.070). Bagging corrects 28 of the 40 test faces the tree gets wrong; on the other 12 both fail. These shared errors are confident, visually plausible confusions, not tie-breaking noise. Bagging fixes variance-driven errors on unstable splits, but not systematic confusions shared by all bootstrap trees.

4. Q2: SAMME Boosting

4.1. Algorithm and numerical safeguards

SAMME [5, 10] is multiclass AdaBoost with hard-label weak learners. With $K = 40$ classes and training weights $w_i^{(1)} = 1/n$, round t fits a tree h_t on the weighted training set, computes

$$\epsilon_t = \sum_{i=1}^{n} w_i^{(t)} \mathbf{1}[h_t(x_i) \neq y_i], \tag{2}$$

and, if $\epsilon_t < 1 - 1/K$, assigns

$$\alpha_t = \log \frac{1-\epsilon_t}{\epsilon_t} + \log(K-1). \tag{3}$$

Weights are updated by

$$w_i^{(t+1)} = \frac{w_i^{(t)} \exp(\alpha_t \mathbf{1}[h_t(x_i) \neq y_i])}{\sum_{j=1}^{n} w_j^{(t)} \exp(\alpha_t \mathbf{1}[h_t(x_j) \neq y_j])}. \tag{4}$$

The ensemble score is $S_k(x) = \sum_{t=1}^{T} \alpha_t \mathbf{1}[h_t(x) = k]$, and the normalised response is

$$q_k(x) = \frac{S_k(x)}{\sum_{t=1}^{T} \alpha_t}, \qquad \sum_{k=1}^{K} q_k(x) = 1. \tag{5}$$

The prediction is $\arg\max_k q_k(x)$.

Two edge cases are explicit. If $\epsilon_t \geq 1 - 1/K$, the learner is no better than random and $\alpha_t \leq 0$; we reject the round, record the reason in `stop_reason`, and stop rather than assign a meaningless weight. If $\epsilon_t \leq \varepsilon$ with $\varepsilon = 10^{-12}$, we clip the error so α_t stays finite. In this run no round was rejected: every requested round produced $\epsilon_t < 0.975$.

4.2. One complete toy round

Table 2 works through a three-class, nine-sample, one-dimensional problem with three samples per class and a depth-1 stump. The stump predicts class 1 for the three class-2 samples, so $\epsilon_1 = 3/9 = 1/3$ and $\alpha_1 = \log(2) + \log(2) = 1.3863$. The six correct samples are multiplied by $e^0 = 1$ and the three wrong samples by $e^{\alpha_1} = 4$; after normalisation the correct weights are $1/18 = 0.0556$

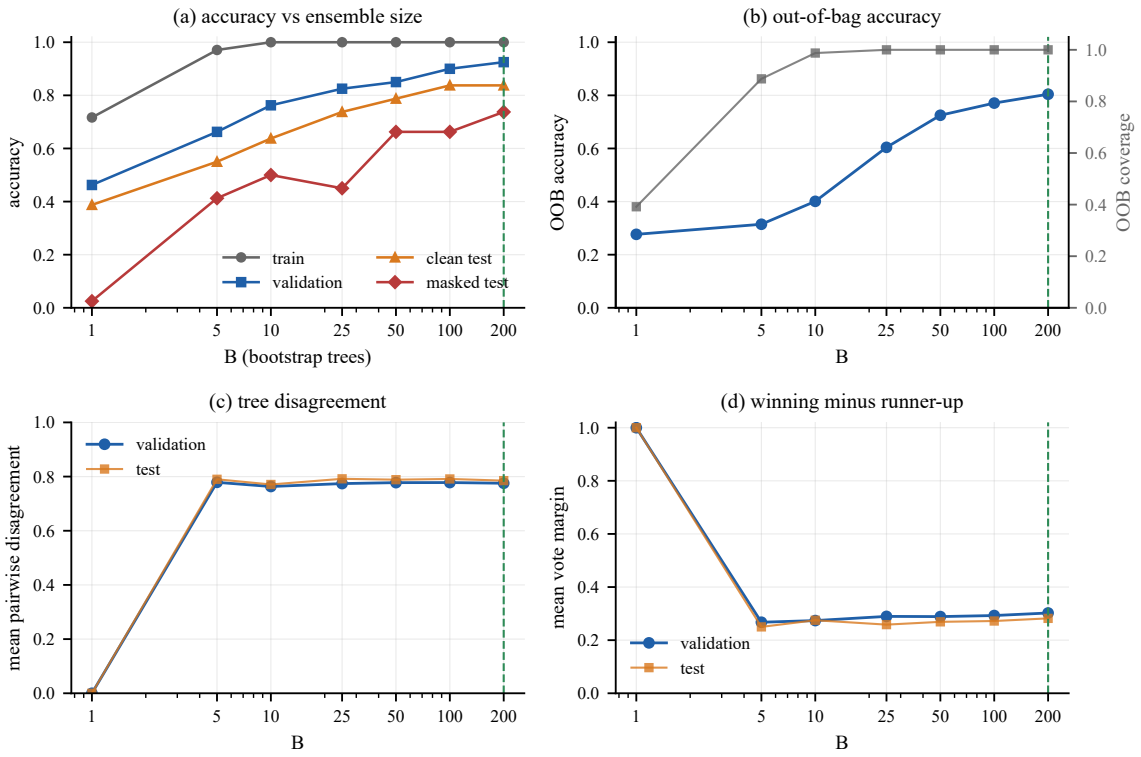


Figure 2. Bagging diagnostics: (a) accuracy versus B; (b) out-of-bag accuracy and coverage; (c) mean pairwise tree disagreement; (d) mean winning-minus-runner-up vote margin. The dashed line marks the selected $B = 200$.

Table 2. One complete SAMME round on a three-class, nine-sample toy problem. The depth-1 stump gives weighted error $\epsilon_1 = 0.3333$ and $\alpha_1 = 1.3863$. Updated weights are normalised.

i	y_i	$w_i^{(1)}$	$h_1(x_i)$	Correct?	$w_i^{(2)}$
0	0	0.1111	0	yes	0.0556
1	0	0.1111	0	yes	0.0556
2	0	0.1111	0	yes	0.0556
3	1	0.1111	1	yes	0.0556
4	1	0.1111	1	yes	0.0556
5	1	0.1111	1	yes	0.0556
6	2	0.1111	1	no	0.2222
7	2	0.1111	1	no	0.2222
8	2	0.1111	1	no	0.2222

and the wrong weights are $2/9 = 0.2222$. The updated weights sum to 1.0000, exactly the behaviour required by Eqs. (3)–(4): hard samples receive four times the weight of easy samples before normalisation.

4.3. Depth × rounds grid

Table 3 summarises the 16-run grid (depths 1–4, requested rounds 10, 25, 50 and 100). All 16 runs accepted every requested round. Depth 1 is too weak: at 100 rounds validation accuracy is only 0.1000. Depth 4 is clearly better, peaking at 50 rounds with 0.4375 validation and 0.3875 test accuracy. More rounds are not monotonically better: 100 rounds at depth 4 gives 0.4250 validation, slightly below

Table 3. SAMME grid. Each cell is clean validation accuracy / actual accepted rounds; the full train/test/timing grid is in `results/boosting_grid.csv`. All 16 runs accepted every requested round.

Depth	10	25	50	100
1	0.037 / 10	0.062 / 25	0.075 / 50	0.100 / 100
2	0.075 / 10	0.138 / 25	0.212 / 50	0.263 / 100
3	0.113 / 10	0.175 / 25	0.287 / 50	0.325 / 100
4	0.237 / 10	0.287 / 25	0.438 / 50	0.425 / 100

50 rounds, while training accuracy is 0.9708 at 50 rounds and the last weighted error is 0.6143. The selected model is therefore depth 4, 50 rounds, not the largest grid cell. Its last α_t is 3.1981, and the 100-round run's final weighted error (0.6694) is higher than the 50-round run's, so additional rounds fit hard faces rather than improving the ensemble.

4.4. Twenty-five, fifty and one hundred learners

Table 4 compares bagging and boosting at the three requested ensemble sizes with frozen settings. Bagging wins at every size: 0.7375, 0.7875 and 0.8375 test accuracy at 25, 50 and 100 trees, versus 0.2875, 0.3875 and 0.3750 for SAMME. Bagging also trains faster in serial-equivalent time at 25 and 50 learners; at 100 the gap narrows because each boosting round refits a depth-4 tree on reweighted data. The comparison is controlled: both families see the same

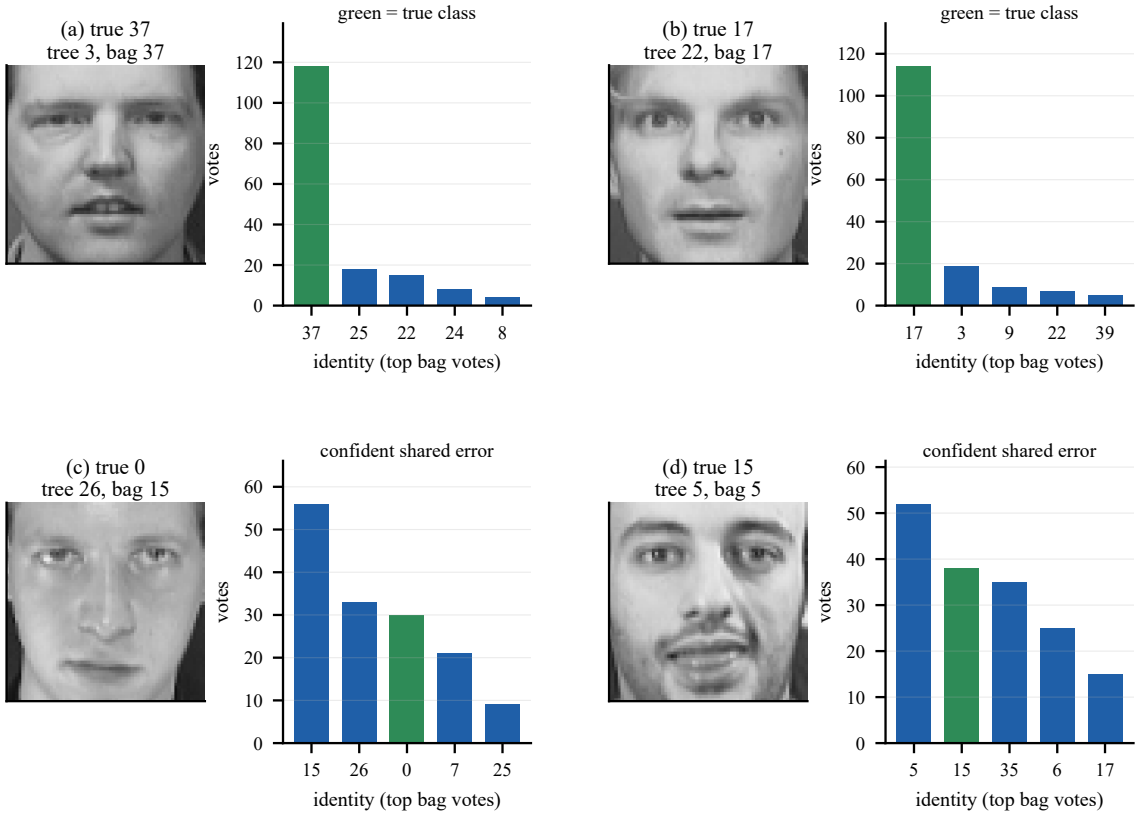


Figure 3. Q1 case studies. Top: faces where bagging corrects the single tree (indices 376 and 175). Bottom: faces where both fail (indices 2 and 154). Bars show the five largest bag vote counts; the true identity is green when present.

Table 4. Controlled comparison at 25, 50 and 100 learners with the frozen settings. Bagging times are serial-equivalent sums of per-tree fit times; boosting is sequential.

Model	Learners	Train	Val	Test	Seconds
Bagging	25	1.0000	0.8250	0.7375	36.0
Boosting	25	0.5208	0.2875	0.2875	13.1
Bagging	50	1.0000	0.8500	0.7875	71.7
Boosting	50	0.9708	0.4375	0.3875	21.2
Bagging	100	1.0000	0.9000	0.8375	146.2
Boosting	100	0.9958	0.4250	0.3750	57.9

240 training images, validation set and test set, and the bagging column uses the same nested bootstrap sequence as Table 1. Its times are sums of per-tree fit times, not wall-clock times distorted by parallel scheduling.

4.5. The face that receives the largest weight

Figure 4 follows dataset index 79 (identity 7), the training face with the largest final SAMME weight. Its final weight is 0.0872, against a uniform weight of 0.00417—about twenty-one times larger. It is misclassified by 44 of the 50 weak learners, yet the cumulative ensemble predicts it correctly from round 1. This is the point of boosting: the face is hard for individual depth-4 trees, so its weight grows, but accumulated votes are already correct and become more so as later learners specialise. Figure 4(b) shows the mechanism: a correct round reduces the weight sharply, but the next tree usually errs again, so the weight climbs back. The top ten training faces hold 0.4481 of the total weight, so reweighting is concentrated.

4.6. Masked response

Figure 5 studies dataset index 170 (true identity 17), a test face that all three clean models classify correctly. Full clean and masked accuracies and mean winning confidences are in Table 5; here we focus on the face where the ensembles diverge. After masking, the single tree predicts 13, bagging predicts 17 with 49 of 200 votes and margin 0.110, and boosting predicts 3. Boosting's normalised score for the true identity is only 0.148, while its score for identity 3 is 0.1807; the top-two margin is 0.032. Bagging's tree disagreement rises from 0.7613 clean to 0.888 masked,

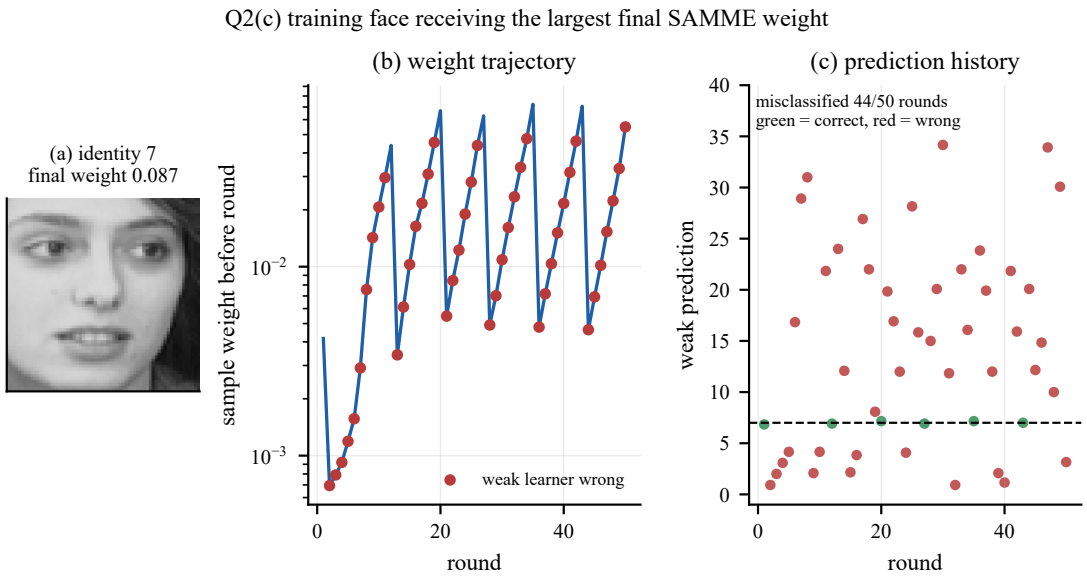


Figure 4. Training face with the largest final SAMME weight (dataset index 79, identity 7). (a) Face and final weight. (b) Weight before each round on a log scale; red points mark rounds where the weak learner is wrong. (c) Weak-learner prediction by round, with the true identity dashed.

so the ensemble is genuinely uncertain, but the true class still leads by 49 votes. The mask removes 256 central pixels, where many identity-specific contours live. Bagging survives because its trees make different errors; boosting's weighted vote is dominated by learners that overfitted to the clean central region. Its reweighted training distribution concentrates on a few hard faces (top ten: 0.4481; heaviest: 0.0872), so later learners specialise on clean central features and have no fallback when that region is removed, whereas bagging still has 49 trees on the true identity.

5. Q3: Measured Bias and Variance

5.1. Estimator and resampling design

For a fixed test image x_i with one-hot target e_i, let $p_r(x_i)$ be the probability vector from refit r and $\bar{p}(x_i) = \frac{1}{R}\sum_{r=1}^{R} p_r(x_i)$. We measure

$$\text{Bias}^2 = \frac{1}{N}\sum_{i=1}^{N}\|\bar{p}(x_i) - e_i\|_2^2, \tag{6}$$

$$\text{Var} = \frac{1}{NR}\sum_{i=1}^{N}\sum_{r=1}^{R}\|p_r(x_i) - \bar{p}(x_i)\|_2^2. \tag{7}$$

This is the assignment's Euclidean probability decomposition [6], not the zero-one loss decomposition [4, 7]; the two need not agree, and accuracy is reported separately.

We draw $R = 30$ within-identity bootstrap training sets. For each identity we sample six of its six training images with replacement, so every resample has 240 images and the same 6/2/2 identity structure as the original split. The selected tree, the 200-tree bag and SAMME (depth 4, 50 rounds) are refitted from scratch and evaluated on the same 80 clean test images. Resample indices are saved in `results/q3_resamples.npz`; per-refit probabilities are in `results/q3_predictions.npz`.

5.2. Selected models

Table 5 and Figure 6(a) give the measured decomposition. The single tree has bias^2 0.4208 and variance 0.6417. Bagging has bias^2 0.5270 and variance 0.0208: variance falls by a factor of 30.9, while squared bias rises by 0.1063. The variance reduction dominates, so accuracy improves. This is the expected trade-off: averaging vote fractions shrinks spread across resamples but also pulls the mean prediction away from a one-hot target. Boosting has bias^2 0.8855 and variance 0.0206. Its variance is also low, but its bias is 2.1 times the tree's, consistent with its low accuracy: depth-4 weak learners are stable but not expressive enough, and hard-vote SAMME does not recover the missing signal. Averaged over 30 refits, the tree's accuracy is 0.4687 ± 0.0490, bagging's is 0.7413 ± 0.0369, and boosting's is 0.2642 ± 0.0680. The ranking matches the single split, though bagging's refit mean (0.7413) is below its single-split estimate (0.8375).

5.3. Depth study

The selected tree is unrestricted, so "deeper than selected" is not defined. We compare a shallow tree (depth 2), a deep finite tree (depth 16) and the selected unrestricted tree (depth

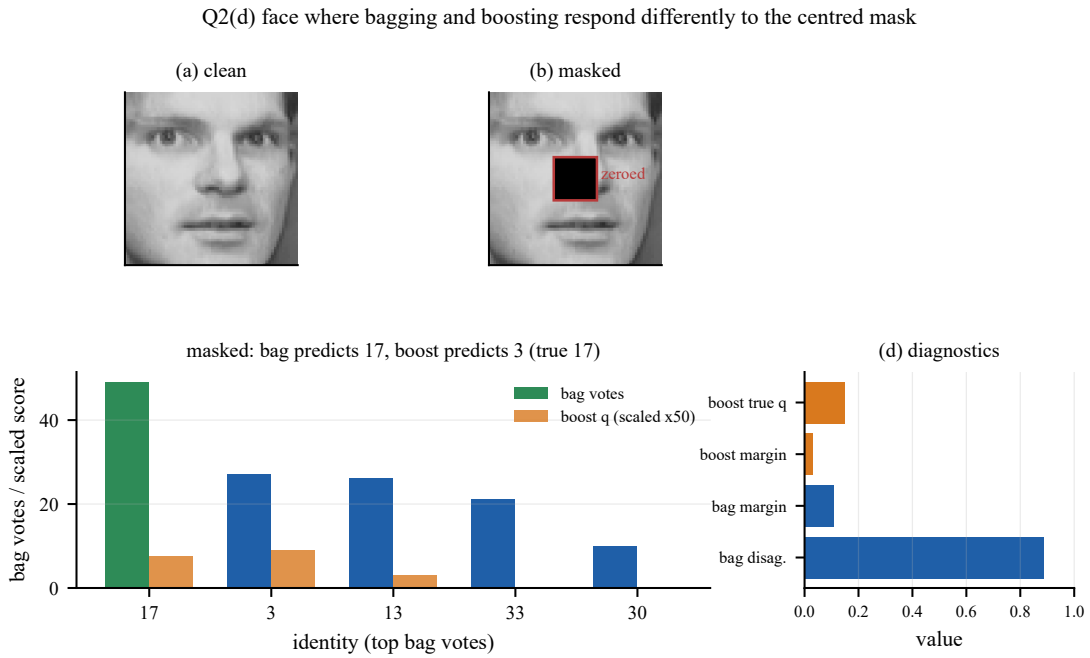


Figure 5. Masked-response case (dataset index 170, true identity 17). (a) Clean image. (b) Image with the centred 16×16 block zeroed. (c) Bag votes and scaled boosting scores for the top five identities. (d) Disagreement and margin diagnostics. Bagging predicts 17; boosting predicts 3.

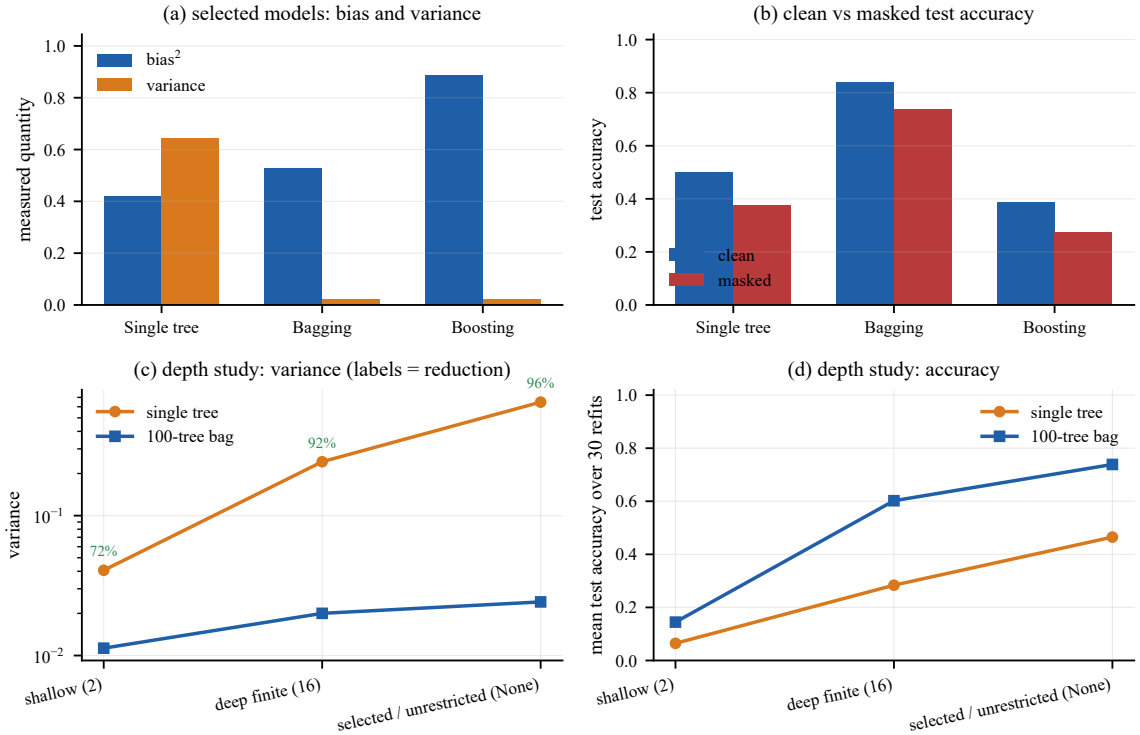


Figure 6. Q3 results. (a) Measured squared bias and variance for the selected models over 30 within-identity resamples. (b) Clean and masked test accuracy. (c) Variance of the single tree and a 100-tree bag at three depths, with percentage reduction printed above each pair. (d) Mean test accuracy over the same 30 refits.

Table 5. Final selected models. Confidence is the mean winning vote/probability; bias^2 and variance are measured over 30 refits. The masked column zeroes the central 16×16 block at test time; training used clean images.

Model	Clean acc.	Masked acc.	Clean conf.	Masked conf.	Bias^2	Var.
Single tree	0.5000	0.3750	1.0000	1.0000	0.4208	0.6417
Bagging	0.8375	0.7375	0.3934	0.2657	0.5270	0.0208
Boosting	0.3875	0.2750	0.1301	0.1372	0.8855	0.0206

`None`), each with a matched 100-tree bag. Figure 6(c,d) gives the results.

Depth 2 has only 3.0 leaves (mean occupancy 80.0, singleton fraction 0.000); it underfits, with bias^2 0.9057, variance 0.0407 and accuracy 0.0646. Its bag raises accuracy to 0.1446, but bias stays at 0.9304. Depth 16 has 18.6 leaves and variance 0.2428; the bag cuts variance by 91.8% and raises accuracy from 0.2838 to 0.6021. The unrestricted tree has 40.0 leaves (mean occupancy 6.0), variance 0.6476 and disagreement 0.823; its bag cuts variance by 96.3%, but raises bias from 0.4224 to 0.5262, and raises accuracy to 0.7387.

The largest accuracy gain is at depth 16(0.318). Thus the answer to the assignment's question is **No**: the deepest tree is not necessarily the depth at which bagging gives the largest accuracy improvement. The deepest tree is where bagging removes the most variance—96.3% versus 91.8% at depth 16 and 72.2% at depth 2—because it has the most leaves and the largest across-resample variance. But the extra variance removal is partly offset by a larger bias increase: bagging raises bias from 0.4224 to 0.5262 at unrestricted depth, compared with 0.6394 to 0.6778 at depth 16.

Disagreement rules out a simple "more diversity is better" explanation. Pairwise disagreement between the 100 bagged trees is highest for shallow trees (0.961), lower at depth 16 (0.885) and lowest at unrestricted depth (0.823). Deep trees converge on the same clean-training partition, so they disagree less with each other, even though their predictions move most when the training set is resampled. Disagreement measures diversity within one ensemble; variance measures sensitivity across resamples. Leaf occupancy explains the other half: depth 2 cannot fit the identities, while the unrestricted tree is most extreme. Depth 16 has enough leaves but a smaller bias penalty from bagging, which is why its accuracy gain is largest. Variance reduction is necessary for the gain, but not sufficient when bias also grows.

Table 5 answers the stability question directly. Boosting has the lowest variance in probability space (0.0206, marginally below bagging's 0.0208), but its refit accuracy is 0.2642 ± 0.0680, so low variance does not imply good predictions: its bias dominates. Bagging has the smallest accuracy standard deviation (0.0369) and the best clean and masked accuracy. The single tree has the largest clean-to-masked drop (0.1250, versus 0.1000 for bagging and 0.1125 for boosting), and is maximally confident in both conditions.

6. Discussion

Bagging wins because the unrestricted tree has relatively low bias and high variance [3]: it represents the training set almost perfectly, but a different bootstrap sample produces different axis-aligned splits. Averaging 200 such trees removes almost all variance (a factor of 30.9) at the cost of a modest bias increase (0.1063), which is why bagging gains 0.3375 test accuracy over the single tree and loses only 0.1000 under the mask.

SAMME is designed for weak learners slightly better than random. With 40 classes the random threshold is 0.975, and a depth-4 tree on 240 faces has weighted error around 0.61, so it is accepted but weak. Reweighting then concentrates on a small set of hard faces (top ten: 0.4481), reducing effective sample size and making the next tree more specialised. The result is stable but biased: variance is low, bias is high, and 100 rounds do not beat 50. Soft probabilities, more data, or stronger weak learners would likely help, but lie outside the frozen protocol.

The mask is a best-case test for ensemble diversity and a worst-case test for models relying on the central face. A single bootstrap tree collapses to 0.0250 masked accuracy, the selected tree drops from 0.5000 to 0.3750, bagging from 0.8375 to 0.7375, and boosting from 0.3875 to 0.2750; the ordering is unchanged, but bagging retains a vote lead while boosting's score flips identity.

Limitations. The test set has only 80 images, so one image is 1.25 accuracy points. All results come from one split and one seed, as required; the 30 resamples estimate refit variability but are not cross-validation. The bias–variance numbers use the Euclidean probability decomposition and are not directly comparable to zero-one decompositions.

7. Conclusion

Under a frozen, roll-number-seeded 6/2/2 split and raw 4096-pixel inputs, bagging is the clear winner: 200 bootstrap trees reach 0.9250 validation and 0.8375 clean test accuracy, and 0.7375 with a centred 16×16 test block zeroed. A single unrestricted tree overfits and reaches 0.5000. A from-scratch SAMME implementation is correct and stable—its toy round matches the closed-form weights, and weighted error falls from 0.8500 to 0.6143—but its hard-vote decisions remain weak at 0.3875. The measured decomposition explains the ranking: bagging cuts variance relative to a matched tree, especially at unrestricted depth, while boosting's low variance is offset by high bias.

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